# Advanced Fixed Income and Credit

Group Assignment

Part I: Deutsche Bank — Finding Relative-Value Trades (HBS 205-059)
Part II: Python Labs — US Treasury Yield Curve Reaction to Trump Tariff Threats

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November 1, 2026



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# 1 Part I — The Deutsche Bank Case Study

### Setup and conventions

Settlement date: **August 15, 2003**. Exhibit 1 lists 52 Treasury coupon bonds whose maturities fall on successive semi-annual dates (Feb 15 / Aug 15) from 2004 to 2029 with no gaps, i.e. exactly one bond at each of the nodes $t = 0.5, 1.0, \ldots, 26.0$ years. This allows the classic *successive substitution* bootstrapping method.

**Compounding convention.** Exhibit 4's footnote states explicitly that Deutsche Bank's own model yields are *Bond-Equivalent Yields* (BEY), i.e. twice the semi-annual compounded yield. Since U.S. Treasuries pay semi-annual coupons, we derive our own zero-coupon yields in the identical BEY convention throughout, so that every comparison against the DB model in Q1.2 is apples-to-apples.

$$\text{BEY} = 2\left(D^{-1/k} - 1\right), \qquad D = \text{discount factor to period } k \text{ (6-month steps)}.$$

## Q1.1 — Bootstrapping the discount curve

**Methodology.** For bond $k$ (maturing at period $k$, coupon $c$ per annum, price $P_k$):

$$P_k = \frac{c}{2}\sum_{j=1}^{k-1} D_j + \left(\frac{c}{2} + 100\right) D_k \quad \Longrightarrow \quad D_k = \frac{P_k - \frac{c}{2}\sum_{j=1}^{k-1} D_j}{\frac{c}{2} + 100}.$$

Solved in order of increasing maturity, using only already-known $D_j$, $j < k$.

**Consistency check.** Our bootstrapped curve matches Deutsche Bank's Exhibit 4 model yields *exactly* at 2Y (1.873%) and 10Y (4.711%). These are two of the maturities to which DB calibrates its three-factor model (the third, one month, lies outside our curve), so an exact match is expected: it confirms that our bootstrap and the BEY convention reproduce the market yields DB uses, rather than validating DB's model itself.

| Maturity (y) | Discount Factor | Zero Yield (BEY) |
|---|---|---|
| 0.5 | 0.99561 | 0.882% |
| 1.0 | 0.98818 | 1.193% |
| 2.0 | 0.96341 | 1.873% |
| 5.0 | 0.84156 | 3.480% |
| 10.0 | 0.62775 | 4.711% |
| 15.0 | 0.42935 | 5.717% |
| 20.0 | 0.30947 | 5.951% |
| 25.0 | 0.23299 | 5.913% |
| 26.0 | 0.22024 | 5.905% |

Table 1: Bootstrapped zero-coupon curve, selected maturities (full 52-point curve in `q1_bootstrapped_curve.csv`).

The curve shape is the standard concave, upward-sloping Treasury curve over the period, flattening from about 15Y onward and a very slight downward drift between 20Y and 26Y.

## Q1.2 — Comparison with Deutsche Bank's model, trading signals

**Logic.** Yield and price move in opposite directions. If our bootstrapped (market-implied) yield is *higher* than DB's model yield, the bond's market price is below the model's "fair" price ⇒ the bond is **cheap** ⇒ **BUY**. If the market yield is *lower*, the bond is **expensive** ⇒ **SELL**. Differences within ±1 bp are labelled "fair".

| Maturity (y) | DB Model Yield | Market Yield (Q1.1) | Diff (bp) | Signal |
|---|---|---|---|---|
| 1 | 1.2443% | 1.1930% | −5.13 | SELL (expensive) |
| 2 | 1.8727% | 1.8727% | +0.00 | fair (DB calibration point) |
| 3 | 2.4110% | 2.4711% | +6.01 | **BUY (cheap)** |
| 4 | 2.9665% | 3.0084% | +4.19 | **BUY (cheap)** |
| 5 | 3.4454% | 3.4799% | +3.45 | **BUY (cheap)** |
| 6 | 3.8557% | 3.8817% | +2.60 | **BUY (cheap)** |
| 7 | 4.1996% | 4.2203% | +2.07 | **BUY (cheap)** |
| 8 | 4.4677% | 4.4865% | +1.88 | **BUY (cheap)** |
| 9 | 4.6528% | 4.6640% | +1.12 | **BUY (cheap)** |
| 10 | 4.7107% | 4.7107% | −0.00 | fair (DB calibration point) |
| 15 | 5.7160% | 5.7167% | +0.07 | fair |
| 20 | 5.9517% | 5.9513% | −0.04 | fair |
| 25 | 5.9315% | 5.9128% | −1.87 | SELL (expensive) |

Table 2: Market vs. DB model comparison and trading signals.

**Pattern.** The belly of the curve (3–9 years) is uniformly *cheap* relative to the DB model, by 1–6 bp, suggesting a BUY signal in that region. The very short end (1Y) and the long end (25Y) look *expensive*. The 2Y, 10Y, 15Y and 20Y points are essentially fair by construction (calibration anchors) or coincidence.

### Q1.3 — Is this risk-free arbitrage?

Because the strategy is directional and model-based, it is **not** a riskless arbitrage, for several reasons:

- **Model risk.** The "mispricing" is only a mispricing relative to Deutsche Bank's three-factor model. If the model is mis-specified (wrong number of factors, wrong functional form, stale calibration), the signal could simply be wrong.
- **Convergence/timing risk.** Even if the model is correctly specified on average, there is no guarantee *when*, or even *whether*, market yields converge back toward the model yields within the trader's holding horizon. The position carries mark-to-market risk.
- **No riskless, capital-structure-free payoff.** A true arbitrage requires a replicating, self-financing portfolio with a knowable, riskless terminal payoff. Here the "trade" is a directional bet on convergence, funded at some financing rate, with basis risk (signal on a Zero Coupon vs. trading on Coupon Bonds) between the position and any hedge.
- **Curve risk while waiting.** Between trade inception and convergence, the *whole* curve can move (parallel shifts, twists), and unless the position is duration/key-rate hedged, those moves can override the small model-implied edge of a few basis points.
- **Implementation costs.** Bid–ask spreads, repo financing (and possible specialness of the bonds shorted) and short-selling constraints can absorb an edge of only 1–6 bp.

In short, this is a **relative-value** trade: a model-informed bet with positive expected value under the trader's own beliefs about the model's validity, not a riskless arbitrage in the classical sense.

### Q2.1 — Cubic spline of the zero-coupon curve

**Methodology.** Natural cubic spline (zero second derivative at the boundary knots) through the five knots specified (2, 5, 10, 15, 20 years), fit directly on the Q1.1 bootstrapped BEY yields. Natural endpoints are an explicit modelling choice, not SciPy's default. Outside $[2, 20]$y the first or last cubic polynomial is continued; the tails are not generally linear.

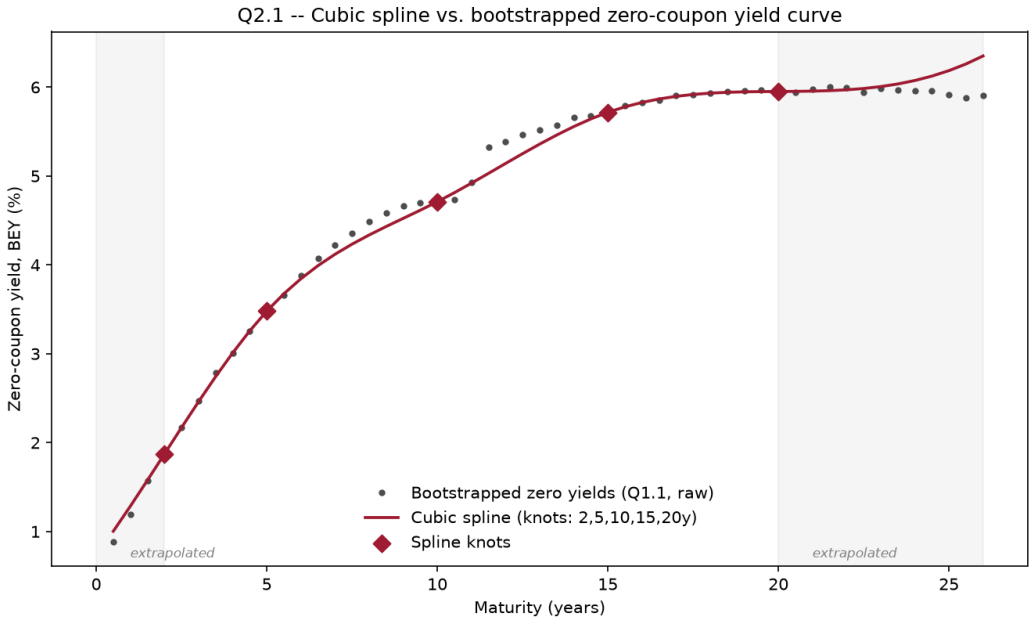



Figure 1: Cubic spline (red) vs. raw bootstrapped zero yields (grey dots). Shaded regions are extrapolated.

**Fit quality.** For the 37 observations within $[2, 20]$y, including the five anchors, the residual (spline minus bootstrap) has mean $-5.50$ bp, sample standard deviation 8.28 bp and maximum absolute value 29.91 bp, attained at 11.5Y with a negative sign. The spline interpolates five selected yields exactly; it does not statistically estimate or average away noise. The high-coupon bonds cited in Exhibit 1 mature at 10.5–12Y, so they cannot determine earlier 6–9Y bootstrap yields in a successive-substitution procedure. The supplied evidence does not establish issuance vintage, liquidity or callability as the cause of the residuals.

## Q2.2 — Does using the spline change the Q1.2 conclusions?

Spline estimates are useful at maturities without an observed point and when a continuous curve is required for valuation or scenario analysis. Replacing available market-implied yields with a five-anchor approximation is a modelling choice; the spline does not exactly reprice all input bonds.

We repeat the Q1.2 comparison using the **spline-implied** yield instead of the bootstrap. The spline must match the five anchors (2,5,10,15,20Y) and may differ elsewhere. The table retains BUY/SELL as *conditional, model-based labels*, with the Q1.2 reporting band: these are comparisons of curve estimates against DB, not new market quotes.

**This matters a great deal.** Six of the thirteen signals (1, 6, 7, 8, 9, 25 years) *flip sign* once the spline replaces the raw bootstrap. Two distinct mechanisms are at work:

1. **Extrapolation risk (1Y and 25Y).** These points are outside $[2, 20]$y. Their spline estimates depend on end-cubic continuation, so the resulting comparisons require explicit tail assumptions; extrapolation alone does not prove that an estimate is invalid.

2. **Interpolation effect (6–9Y).** At 6, 7, 8 and 9Y, the spline lies respectively 3.78, 10.08, 15.03 and 14.05 bp below the bootstrap. These differences exceed the corresponding positive bootstrap-minus-DB gaps and reverse their signs. This is an effect of the five-anchor interpolation, not evidence that later high-coupon bonds caused noise in the earlier bootstrap yields. The bootstrap is almost flat between 9 and 11Y and then jumps at 11.5Y (Figure 1); the 10Y knot (4.711%) sits at the bottom of this 10.5–11.5Y step, so the spline sags below the bootstrap between the 5Y and 10Y knots.

| Maturity (y) | DB Model | Spline Yield | Diff (bp) | Spline Signal | *Changed vs. Q1.2?* |
|---|---|---|---|---|---|
| 1 | 1.2443% | 1.2846% | +4.03 | BUY | **YES** (was SELL) |
| 2 | 1.8727% | 1.8727% | +0.00 | fair | no (knot) |
| 3 | 2.4110% | 2.4608% | +4.98 | BUY | no |
| 4 | 2.9665% | 3.0096% | +4.31 | BUY | no |
| 5 | 3.4454% | 3.4799% | +3.45 | BUY | no (knot) |
| 6 | 3.8557% | 3.8439% | −1.18 | SELL | **YES** (was BUY) |
| 7 | 4.1996% | 4.1195% | −8.01 | SELL | **YES** (was BUY) |
| 8 | 4.4677% | 4.3363% | −13.14 | SELL | **YES** (was BUY) |
| 9 | 4.6528% | 4.5235% | −12.93 | SELL | **YES** (was BUY) |
| 10 | 4.7107% | 4.7107% | −0.00 | fair | no (knot) |
| 15 | 5.7160% | 5.7167% | +0.07 | fair | no (knot) |
| 20 | 5.9517% | 5.9513% | −0.04 | fair | no (knot) |
| 25 | 5.9315% | 6.1859% | +25.44 | BUY | **YES** (was SELL) |

Table 3: Q1.2 signals recomputed off the Q2.1 spline curve.

**Conclusion.** The comparison is sensitive to curve construction. Substituting spline estimates for market-implied yields changes the object being compared with DB. The six reversals do not establish that the bootstrap is erroneous, that the spline removes noise or that a spline-based SELL is superior to the original BUY candidate. Trading conclusions require consistent cash-flow pricing and an assessment of model risk and costs.

## Q3.1 — Nelson–Siegel–Svensson fit

**What is estimated.** The question asks for the *discount function*. We estimate it in spot-rate space and map back, keeping the Q1.1 bond-equivalent-yield convention throughout so that every comparison with the bootstrap is apples-to-apples:

$$D(m) = \left(1 + \tfrac{y(m)}{2}\right)^{-2m}, \qquad y(m) = \beta_0 + \beta_1\Lambda_1(m) + \beta_2\Lambda_2(m) + \beta_3\Lambda_3(m),$$

$$\Lambda_1(m) = \frac{1 - e^{-m/\tau_1}}{m/\tau_1}, \qquad \Lambda_2(m) = \Lambda_1(m) - e^{-m/\tau_1}, \qquad \Lambda_3(m) = \frac{1 - e^{-m/\tau_2}}{m/\tau_2} - e^{-m/\tau_2}.$$

Setting $\beta_3 = 0$ gives the three-factor Nelson–Siegel (NS) model; retaining it gives the four-factor Svensson (NSS) extension. $\Lambda_1$ decays monotonically from 1 to 0, so $\beta_1$ is a short-end (slope) factor; $\Lambda_2$ is zero at both ends and peaks at $m^\star = 1.7933\,\tau_1$, so $\beta_2$ bends only the belly; the loading on $\beta_0$ is constant, so $\beta_0 = y(\infty)$ is the asymptotic long rate.

**Methodology.** Estimating all six parameters $(\beta_0, \ldots, \beta_3, \tau_1, \tau_2)$ jointly by unconstrained non-linear least squares proved numerically fragile: different starting points converged to different, equally-well-fitting local optima. The cause is a near-linear dependence among the loadings once $\tau_1$ is large relative to the sample. Regressing the curvature loading on the other two over our 0.5–26y grid:

| $\tau_1$ | $R^2$ of $\Lambda_2$ on $[1, \Lambda_1]$ | cond$(X)$ | corr$(\Lambda_1, \Lambda_2)$ |
|---|---|---|---|
| 2.00 y | 0.389 | 19.0 | +0.62 |
| 5.00 y | 0.099 | 22.5 | −0.31 |
| 12.08 y | **0.905** | **60.8** | **−0.95** |
| 20.00 y | 0.976 | 139.5 | −0.99 |

At $\tau_1 \approx 12$y the curvature loading is over 90% explained by a linear combination of the level and slope loadings, so the three betas are not separately identified.

We therefore **concentrate the betas out of the objective** (profiled, or variable-projection, least squares). Conditional on $(\tau_1, \tau_2)$ the loadings are fixed basis functions of maturity, the betas

enter linearly, and the inner problem is an OLS regression with a unique closed-form solution; only the low-dimensional $\tau$ problem is searched, on a grid of $\tau_1 \in [0.3, 25]$y (500 points, step 0.049y) for the three-factor model and a $80 \times 80$ grid over $\tau_1 \in [0.3, 15]$y, $\tau_2 \in [2, 30]$y for Svensson, excluding $|\tau_1 - \tau_2| < 0.5$y to avoid the degenerate $\tau_1 \simeq \tau_2$ corner. This is the estimator of Nelson & Siegel (1987), systematised by Gilli, Große & Schumann (2010). Diebold & Li (2006) instead *fix* $\lambda = 1/\tau_1 = 0.0609$ a priori — placing the curvature peak at 30 months, i.e. $\tau_1 = 1.37$y — so that no search is needed at all. We borrow that device in the constrained variant below, which restricts $\tau_1$ to $[1.0, 3.5]$y, a band centred on the Diebold–Li value.

**Estimation choices (documented).**

1. **Objective.** Equally weighted squared errors on *zero-coupon yields*, not on prices. Yield-space weighting treats every maturity alike; price-space weighting would let the long end dominate through its duration and would effectively re-weight towards the 10.5–12Y high-coupon bonds cited in Exhibit 1.
2. **Convention.** BEY (semi-annual) throughout, matching Q1.1–Q1.2.
3. **Sample.** All 52 bootstrapped nodes, $m = 0.5, \ldots, 26$y. No outlier is removed: the 10.5–11.5Y step is left in so that its effect on the fit is visible.
4. $\tau$ **selection.** Profiled RSS minimisation over the full grid, *plus* a constrained variant with $\tau_1$ restricted to $[1.0, 3.5]$y — the conventional short-end range, bracketing the Diebold–Li value $\tau_1 = 1.37$y. Both are reported, for the reason given under "Identification" below.
5. **Factor count.** Three versus four, decided below.

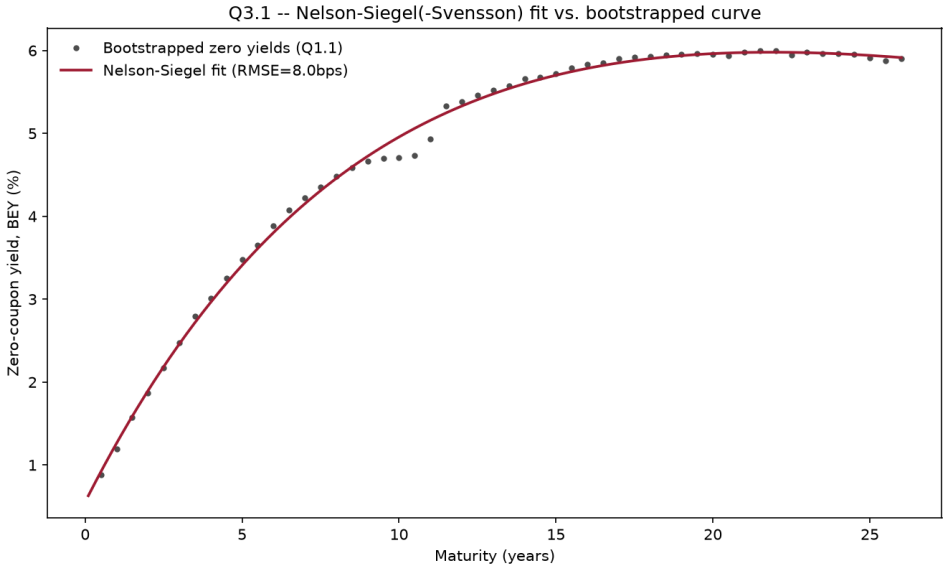



Figure 2: *Profiled* Nelson–Siegel fit ($\tau_1 = 12.08$y, RMSE 7.96 bp) vs. the Q1.1 bootstrapped curve. The identified fit of Table 4 is not plotted.

**Identification: the profiled optimum is not economically interpretable.** The profiled RSS surface is nearly flat in $\tau_1$, while the betas move enormously along that ridge:

| $\tau_1$ | $\beta_0$ (level) | $\beta_1$ (slope) | $\beta_2$ (curvature) | RMSE |
|---|---|---|---|---|
| 2.0 y | 6.951% | −6.413% | −3.871% | 12.20 bp |
| 5.0 y | 6.730% | −6.506% | +3.445% | 12.08 bp |
| 9.0 y | 4.254% | −3.817% | +10.982% | 8.58 bp |
| 12.08 y (profiled optimum) | 0.695% | −0.140% | +17.926% | 7.96 bp |

The profiled optimum does fit better — by 4.1 bp of RMSE against the constrained fit with $\tau_1 \in [1.0, 3.5]$y — but it buys that fit by abandoning the factor interpretation entirely. Taken at face value it states that the asymptotic long rate is 0.695%, against a bootstrapped 26Y yield of 5.905%, and that the slope factor, which must account for the 500 bp rise from the short end to the long end, is −14 bp. Neither is an economically meaningful magnitude.

Against that, 4.1 bp has to be judged on the scale of the errors this fit already carries: the maximum residual is 32.5 bp at 10.5Y, in the 10.5–11.5Y step that neither smooth model reproduces. The extra fit bought by a free $\tau_1$ is therefore of the same order as a feature neither functional form can represent in any case, while the cost — a decomposition that means nothing — is not. We report both fits and carry out the Q3.2 decomposition on the identified one.

**Model selection: three or four factors?** The plain three-factor NS model achieves RMSE = 7.96 bp; the four-factor Svensson improves this to 7.28 bp — but it does so with

$$\beta_0 = -13.938\%, \quad \beta_1 = +14.302\%, \quad \beta_2 = +8.837\%, \quad \beta_3 = +54.892\%,$$
$$\tau_1 = 4.022\text{y}, \quad \tau_2 = 16.886\text{y}.$$

Formal tests favour the larger model: a nested $F$-test rejects NS at the 5% level ($F(2, 46) = 4.52$, $p = 0.016$), and both AIC (−734.1 vs. −739.4) and BIC (−726.3 vs. −727.7) prefer Svensson.

We adopt the three-factor model regardless, because the Svensson parameter vector is not a description of a second hump. Its asymptotic long rate is *negative* (−13.94%), held up by an offsetting $\beta_1$ of +14.30%; and its second curvature factor carries a loading of 54.89% with $\tau_2 = 16.886$y, placing that hump's peak at $1.7933\,\tau_2 = 30.3$ years — four years beyond the longest bond in the sample. The fourth factor is thus a second *out-of-sample* level factor, not a second hump: it is the same identification pathology as above, now with two decay parameters instead of one. The 0.68 bp it buys is also small next to the 29.9 bp maximum spline residual reported at 11.5Y in Q2.1. The $F$-test is measuring in-sample flexibility, not economic content.

| Parameter | Profiled fit ($\tau_1$ free) | Identified fit ($\tau_1 \in [1.0, 3.5]$y) |
|---|---|---|
| $\beta_0$ (level) | 0.695% | 6.913% |
| $\beta_1$ (slope) | −0.140% | −6.260% |
| $\beta_2$ (curvature) | 17.926% | −4.598% |
| $\tau_1$ | 12.081 y | 1.834 y |
| Curvature peak $1.7933\,\tau_1$ | 21.7 y | 3.3 y |
| RMSE | 7.96 bp | 12.10 bp |
| Implied $y(\infty) = \beta_0$ | 0.695% | 6.913% |
| Implied $y(0) = \beta_0 + \beta_1$ | 0.554% | 0.654% |

Table 4: Nelson–Siegel parameters: profiled optimum vs. identified fit. Bootstrapped reference points: $y(0.5) = 0.882\%$, $y(26) = 5.905\%$.

**Fit quality and the implied discount function.** Mapping the *profiled* fitted yields back through $D(m) = (1 + y(m)/2)^{-2m}$:

| $m$ (y) | $y_{\mathrm{NS}}(m)$ | $D_{\mathrm{NS}}(m)$ | $D_{\mathrm{boot}}(m)$ | diff |
|---|---|---|---|---|
| 0.5 | 0.918% | 0.99543 | 0.99561 | −0.00018 |
| 1.0 | 1.262% | 0.98750 | 0.98818 | −0.00068 |
| 2.0 | 1.895% | 0.96298 | 0.96341 | −0.00043 |
| 5.0 | 3.409% | 0.84451 | 0.84156 | +0.00295 |
| 10.0 | 4.957% | 0.61284 | 0.62775 | **−0.01492** |
| 15.0 | 5.702% | 0.43029 | 0.42935 | +0.00094 |
| 20.0 | 5.962% | 0.30880 | 0.30947 | −0.00067 |
| 25.0 | 5.941% | 0.23141 | 0.23299 | −0.00158 |
| 26.0 | 5.915% | 0.21969 | 0.22024 | −0.00055 |

The residual is within about 7 bp everywhere except around the 10.5–11.5Y step, where the fitted yield sits 24.6 bp *above* the bootstrap at 10Y and 32.5 bp above at 10.5Y — a discount factor wrong by about 1.5 cents per dollar, the largest error in the fit. This is the *same* 10.5–11.5Y step that generated the largest spline residuals in Q2.1. Two curve models of entirely different construction — a local interpolant and a global three-parameter parametric form — both fail to reproduce it, which shows the step is not a shape either functional form can represent. It does not establish what causes it: as Q2.1 notes, the supplied evidence does not identify issuance vintage, liquidity or callability. Nor does it bear on the 6–9Y signals of Q1.2, which Q2.2 attributes to the spline's own geometry — the 10Y knot sits at the bottom of the step and pulls the spline down between 5Y and 10Y.

### Q3.2 — Factor loadings and economic drivers

**The loadings.** The three loadings are, by construction, orthogonal in *shape* rather than in value: $\Lambda_0 \equiv 1$ is flat and shifts every maturity equally; $\Lambda_1$ falls monotonically from 1 at $m = 0$ to 0 at $m = \infty$ and therefore moves only the short end; and $\Lambda_2$ is a hump, zero at both ends, peaking at

$$m^\star = 1.7933\,\tau_1,$$

so it tilts the belly without disturbing either asymptote. This is what licenses the level, slope and curvature reading, and it holds only when $m^\star$ sits comfortably *inside* the sample.

**Why the profiled decomposition cannot be interpreted.** With $\tau_1 = 12.08$y the curvature loading peaks at $m^\star = 21.7$ years and has fallen back only to 0.294 by $m = 26$ (it reaches 0.12 at 100y and 0.02 at 500y). Over the *observed* range it is therefore monotone increasing and near-flat at the long end — visually and statistically a second level/slope factor rather than a hump. The regression reported in Q3.1 quantifies this: $\Lambda_2$ is 90.5% explained by $[1, \Lambda_1]$. The least-squares step consequently loads almost all the explanatory power onto $\beta_2$ (17.9%), leaving a level of 0.695% and a slope of −0.140% that bear no relation to the curve's actual asymptote (5.9%) or actual slope (+500 bp). The profiled fit is an excellent *interpolant* and a meaningless *decomposition*; Figure 3 shows exactly this. We therefore decompose the identified fit of Table 4.

**Economic drivers.** Settlement is 15 August 2003, and each factor maps onto a distinct macro-financial driver observable on that date.

- **Level ($\beta_0 = 6.913\%$) — long-run nominal anchor.** The asymptotic rate is the sum of expected long-run real growth, expected long-run inflation and a term premium. These move mainly with long-run expectations rather than with the business cycle, which is why the loading is flat: a repricing of trend inflation moves the entire curve. Note that 6.91% sits about 100 bp above the longest yield we observe (5.905% at 26Y): with $\tau_1$ short, the model extrapolates the still-rising mid-curve out to infinity. $\beta_0$ should therefore be read as the model's long-run anchor, not as a rate quoted anywhere in the market.
- **Slope ($\beta_1 = -6.260\%$) — monetary policy stance.** $\beta_1 = y(0) - y(\infty)$, so a large negative $\beta_1$ is a very steep curve. In August 2003 the Fed funds target stood at 1.00%

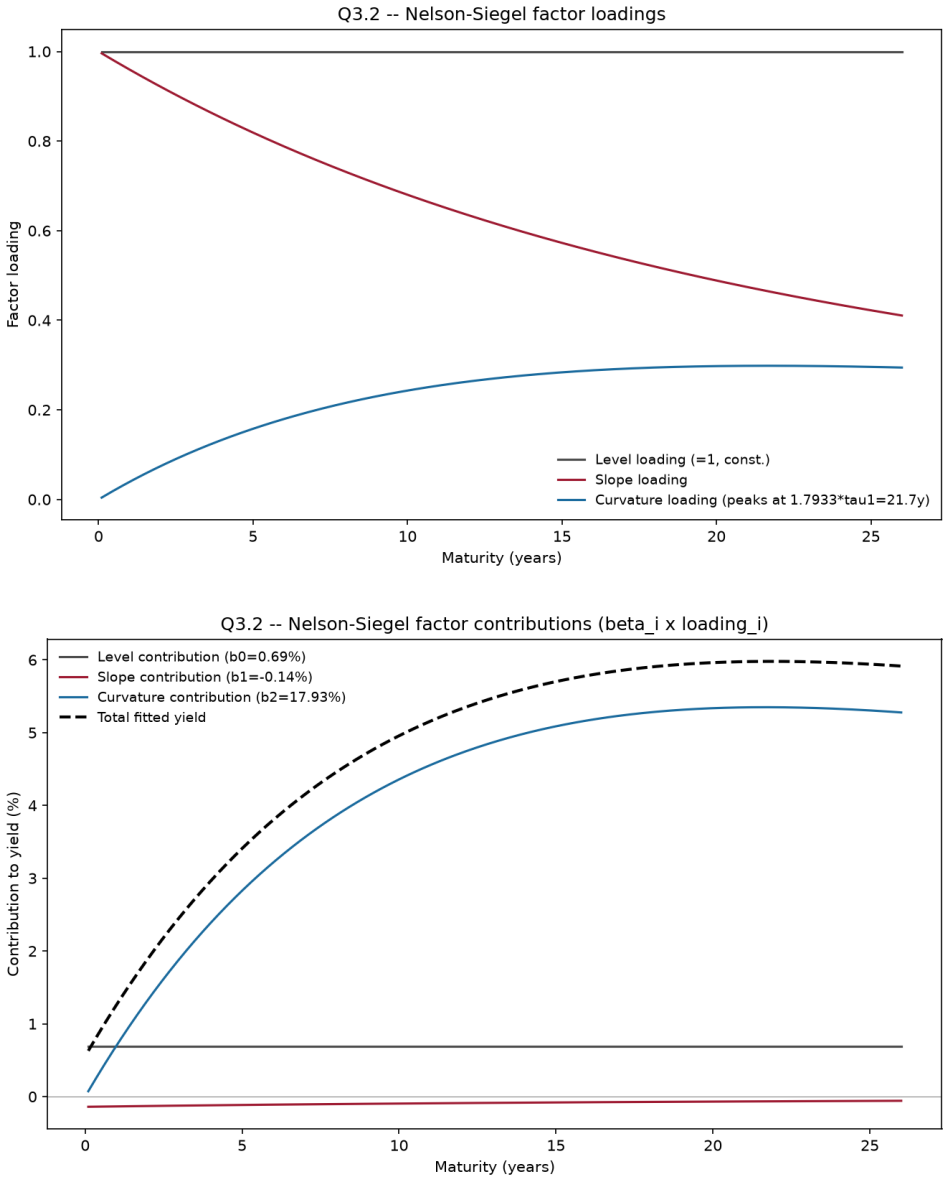



Figure 3: Loadings (top) and contributions $\beta_i \times \Lambda_i(m)$ (bottom) of the *profiled* fit, $\tau_1 = 12.08$y. The curvature loading does not peak until $1.7933\,\tau_1 = 21.7$y, so over $[0, 26]$y it is monotone and absorbs the level: the grey level contribution is a flat 0.695% and the red slope contribution is indistinguishable from zero.

after the June 2003 cut, the trough of the post-dot-com easing cycle and the lowest policy rate in 45 years, set against an explicit deflation scare earlier that year. The fit implies $y(0) = \beta_0 + \beta_1 = 0.654\%$, within 23 bp of our bootstrapped 6-month yield of 0.882% and consistent with a 1% policy rate. The slope factor is therefore measuring the gap between an emergency-low policy rate and an unchanged long-run anchor: 626 bp, which combines expected normalisation with a term premium.

- **Curvature ($\beta_2 = -4.598\%$, peaking at 3.3y) — the expected path, not the endpoints.** Curvature captures the *timing* of the normalisation priced by the slope factor, and its sign here is informative: $\beta_2 < 0$ pushes the 2–5y sector *below* the path implied by level and slope alone, i.e. the market expects the Fed to stay low first and move later, rather than to normalise at a constant exponential rate. That is precisely the guidance in force at settlement: on 12 August 2003, three days before our valuation date, the FOMC held the funds target at 1% and stated that policy accommodation could be maintained "for a considerable period" — language that Fed funds futures read as a commitment not to tighten for at least six months. A delayed-liftoff path depresses the front belly exactly where $\Lambda_2$ loads most heavily.

  The same sector is where convexity flows were concentrated: between mid-June and mid-August 2003 the 10Y yield rose roughly 150 bp as mortgage servicers sold duration into a falling market. Our largest residuals (+32.5 bp at 10.5Y) sit just beyond that region, in the 10.5–11.5Y step that no smooth model reproduces; we therefore read the curvature factor as the policy-path signal and leave the step's cause open.

**Relation to Deutsche Bank's model.** This is the same decomposition logic as DB's three-factor model in Exhibit 3, and it explains why DB calibrates to the one-month, 2Y and 10Y points (Q1.1): three maturities pin down level, slope and curvature. It also explains the Q1.2 pattern. A three-factor model cannot reproduce idiosyncratic, bond-specific deviations by construction, so any residual between DB's model and the market is either genuine relative value or bond-specific pricing — and on the evidence available neither Q2.2 nor the 10Y discount-factor residual above settles which of the two applies in the 6–9Y sector.

**Caveat carried forward.** The identification problem of Q3.1 is not an artefact of our estimator; it is a property of fitting a three-factor exponential model on a sample whose maturity range is not much longer than $\tau_1$. When $m^\star$ approaches the long end of the data, the "curvature" factor and a slow-moving long-rate factor become observationally equivalent. Any practitioner tracking NS betas through time must therefore fix $\tau_1$ (as Diebold & Li do) rather than re-optimise it each period, or the resulting beta series will be contaminated by movements in $\tau$ masquerading as movements in level and curvature.

## Q4 — Answer summary

**Q4.1.** A +50 bp bump of the 10Y knot with the natural spline re-fitted raises the 10Y zero yield by exactly 50 bp and leaves the other four knots unchanged, but the shock is *not* local: the curve ripples (+22.1 bp at 7Y, −8.4 bp at 17Y, −11.5 bp at 26Y).

**Q4.2.** $\text{KRD}_{10Y} = 2.21$ years ($\Delta P = -\$10{,}813.14$, i.e. −1.08% of NAV; DV01 = \$221.0 central, \$216.3 one-sided).

**Q4.3.** The 52 individual zero-yield KRDs sum to 7.981 years, exactly the portfolio's effective duration; the aggregated key-rate profile is 0.37/1.54/2.21/0.81/3.06 years at 2/5/10/15/20Y (the 15Y/20Y split depends on the extrapolated tail: 1.85/2.09 under a flat tail beyond 20Y).

## Q4.1 — Shocking the 10Y key rate

**Methodology.** The Q2.1 curve is a natural cubic spline through the bootstrapped zero yields at the knots 2, 5, 10, 15 and 20y, which are exactly the assignment's key rates. We therefore

implement "shock the 10-year rate by 50 bp" literally on that model: we add +50 bp to the 10Y knot value, *re-fit* the natural spline, and define the curve shock as

$$\Delta y(t) = S_{\text{bumped}}(t) - S_{\text{base}}(t), \qquad t = 0.5, 1.0, \ldots, 26\text{y}.$$

Because spline interpolation is linear in the knot values, $\Delta y(t)$ does not depend on the level of the curve, and the five key-rate shocks sum *exactly* to a +50 bp parallel shift at every maturity (verified in the code). For comparison, we also run the textbook Ho (1992) triangular ("tent") shock: full size at its own knot, decaying linearly to zero at the adjacent knots, and held flat beyond the outer knots.

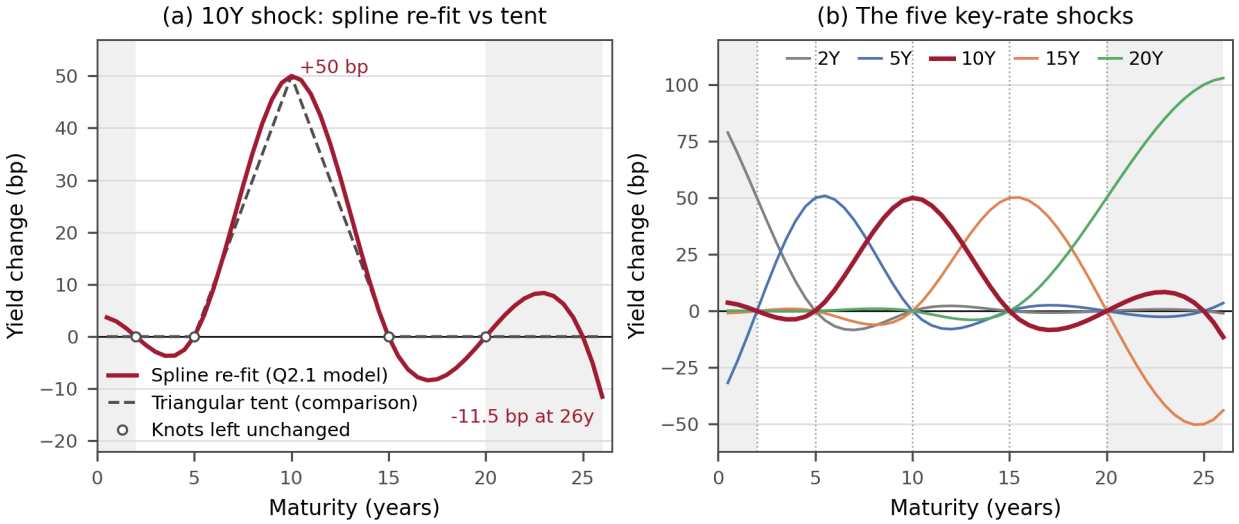



Figure 4: Left: the curve shock $\Delta y(t)$ from re-fitting the spline (red) vs. the triangular tent (dashed); hollow markers are the four knots left unchanged. Right: the five spline key-rate shocks. Shaded areas: extrapolated maturities ($< 2$y, $> 20$y).

**Impact on the curve.** The shock is +50 bp at 10Y and, by construction, zero at the other knots (2, 5, 15, 20y). However, unlike the tent, the spline response is **not local**:

- Between 5Y and 15Y it is *wider* than the tent: +22.1 bp at 7y and +37.0 bp at 12y, versus +20 and +30 bp for the linear ramp.
- Outside $[5, 15]$y the cubic *ripples* with alternating signs: about −3 bp between 3 and 4y, −8.4 bp at 17y, and +8.4 bp at 23y.
- In the **extrapolated** region beyond 20y the spline continues the last cubic polynomial — the `CubicSpline` implementation extends the end cubic, it does not hold the curve flat — so the ripple grows to −11.5 bp at 26y. (A flat extension would put the 26Y yield at 5.95% instead of 6.35% and would set this last contribution to zero; we keep the model-consistent extension and quantify the sensitivity in Q4.3.) The right panel shows that this effect is much stronger for the outer knots: a 20Y bump moves the 26y yield by about +103 bp, a 15Y bump moves it by about −44 bp, and a 2Y bump moves the 0.5y yield by +79 bp.

| Maturity (y) | 2 | 3 | 5 | 7 | 10 | 12 | 15 | 17 | 20 | 23 | 26 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $\Delta y$, spline re-fit (bp) | 0.0 | −2.9 | 0.0 | +22.1 | +50.0 | +37.0 | 0.0 | −8.4 | 0.0 | +8.4 | −11.5 |
| $\Delta y$, tent (bp) | 0.0 | 0.0 | 0.0 | +20.0 | +50.0 | +30.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |

Table 5: The +50 bp 10Y shock at selected maturities: global spline re-fit vs. the local triangular tent. $\Delta y$ is zero at 5Y and 15Y because the tent decays to zero at the adjacent knots; it is zero at 2, 5, 15 and 20Y for the re-fit because those knots are not moved.

In short, a key-rate shock is only as local as the curve model it is applied to. The tent is local

by design, while a cubic spline spreads a single-node shock across the whole curve, most strongly where the curve is extrapolated rather than interpolated.

### Q4.2 — Portfolio KRD to the 10Y shock

**Portfolio construction (assumption).** "Equally invested" is read as an equal *dollar market value*: \$1,000,000/52 ≈ \$19,230.77 in each bond, i.e. a face amount of (dollar amount/price) × 100 per bond. The alternative reading, an equal *face* amount (\$19,230.77 of face in each bond), would give a \$1.19m market value and $\mathrm{KRD}_{10Y} = 2.31$y: with equal face, long bonds trading above par (10.5–20y, prices 130–174) get more market-value weight than near-par short bonds (about 103), shifting weight towards the 10Y bucket and raising duration. We use equal market value, the standard reading of "equally invested", and report the alternative in the robustness table.

**Approach.**

1. Build the 52 × 52 cash-flow matrix (semi-annual coupons plus principal) and aggregate it into the portfolio cash flow at each semi-annual date.
2. *Base valuation* uses the Q1.1 bootstrapped zero curve (BEY, semi-annual compounding). It reprices every bond exactly, so the base NAV is exactly \$1,000,000. Pricing off the spline itself would give \$1,002,389, and the +\$2,389 gap (spline NAV − bootstrap NAV, the Q2.2 fitting residual) would contaminate the risk measure.
3. *Shocked valuation*: add the spline-implied $\Delta y(t)$ from Q4.1 to the bootstrapped yields and reprice. The shape of the shock therefore comes from the Q2.1 model, while the base level matches market prices.
4. Report the result as (i) the dollar P&L of the +50 bp shock; (ii) the KRD in years, using a central difference to strip out convexity, $\mathrm{KRD}_k = -\dfrac{P(+h) - P(-h)}{2\,P_0\,h}$ with $h = 50$ bp; and (iii) the DV01 (\$ per 1 bp), which is the quantity used to size hedges.

**The residual is held fixed.** With $s(t)$ the Q2.1 spline, $b(t)$ the Q1.1 bootstrap and $r(t) = b(t) - s(t)$, pricing the base at $b$ and the shocked book at $b + \Delta y$ is equivalent to assuming that $r(t)$ does not move when the knot does: the shock shifts the model curve and the bond-specific residual is carried unchanged. The assumption is deliberate — it keeps $P_0$ at the \$1m actually invested and measures the sensitivity of that mark — but it is an assumption. If the residual is a liquidity or rich–cheap premium that itself responds to the curve, the true exposure differs. The pure-model alternative ($r \equiv 0$, repricing off the spline) gives 2.234y, 1.1% higher (robustness table below); the residual is small relative to a 50 bp shock, which is why the choice barely matters.

**Result.**

$$\begin{aligned} \Delta P_{10Y} &= \$989{,}186.86 - \$1{,}000{,}000.00 = -\$10{,}813.14 \ (-1.08\% \text{ of NAV}), \\ \mathrm{KRD}_{10Y} &= 2.21 \text{ years}, \qquad \mathrm{DV01}_{10Y} = \$221.0 \end{aligned}$$

The sign is as expected: a long bond portfolio loses value when yields rise. The \$221.0 is a *central-difference* DV01, $\mathrm{DV01}_{10Y} = -\dfrac{P(+h) - P(-h)}{2h} \times 10^{-4} = \mathrm{KRD}_{10Y} \times P_0 \times 10^{-4}$ with $h = 0.005$; the *one-sided* +50 bp P&L implies \$216.26 per bp, so $\Delta P \neq -50 \times \mathrm{DV01}$. The \$4.8 per bp gap is the convexity term, which a 50 bp move is large enough to reveal; we report the central figure because it is the one used to size the hedge in Q4.3.

**What actually moves the number (robustness).** The KRD is essentially invariant to the pricing engine and materially sensitive to the *shock design*, which is where most of the dispersion between reasonable implementations comes from:

| Variant | $\Delta P$ for +50 bp | $\text{KRD}_{10Y}$ (yrs) |
|---|---|---|
| Spline re-fit, bootstrapped base (**reported**) | −\$10,813 | 2.210 |
| Spline re-fit, spline-priced base | −\$10,954 | 2.234 |
| Triangular tent, bootstrapped base | −\$10,123 | 2.061 |
| Triangular tent, spline-priced base | −\$10,232 | 2.078 |
| Spline re-fit, linear extension outside $[2, 20]$y | −\$12,279 | 2.511 |
| Single-point 10Y bump (10Y zero yield only) | −\$1,169 | 0.240 |
| Equal-face portfolio, bootstrapped base | −\$13,445 | 2.307 |

Every knot-based convention puts the 10Y exposure between 2.0 and 2.5 years (2.06–2.51y); the spread is methodological, not arithmetic. We report 2.21y because (i) Q4.1 asks for the impact on the curve *estimated in Q2.1*, which is the re-fitted spline, and (ii) pricing the base off the bootstrap keeps $P_0$ exactly at the \$1m actually invested, so the +\$2,389 Q2.2 fitting residual (spline NAV − bootstrap NAV) cannot leak into the risk measure.

**Two senses of "key rate".** A key-rate duration is the sensitivity to a one-unit move in one coordinate of the curve. The coordinate can be a knot of a curve model, in which case the shock drags neighbouring yields with it (2.21y here), or a single zero yield, in which case only that discount factor moves (0.24y). Q4.3(a) computes the second object for all 52 dates, so the 0.24y row is its 10Y entry, not an error: the two numbers differ by a factor of nine because the knot shock also moves the 5–15y curve.

## Q4.3 — Full key-rate duration profile

**(a) Each of the 52 zero-coupon yields individually (the literal reading).** "All maturities" is read as applying the same +50 bp shock to each semi-annual zero yield (0.5y to 26y) *on its own*, leaving the other 51 unchanged. This version needs no curve model: only the discount factor of that single date moves, so to first order the KRD at date $t$ is $t \times \text{PV}(\text{CF}_t)/\text{NAV}$ (i.e. $[t/(1 + y_t/2)] \times \text{PV}(\text{CF}_t)/\text{NAV}$ in BEY terms).

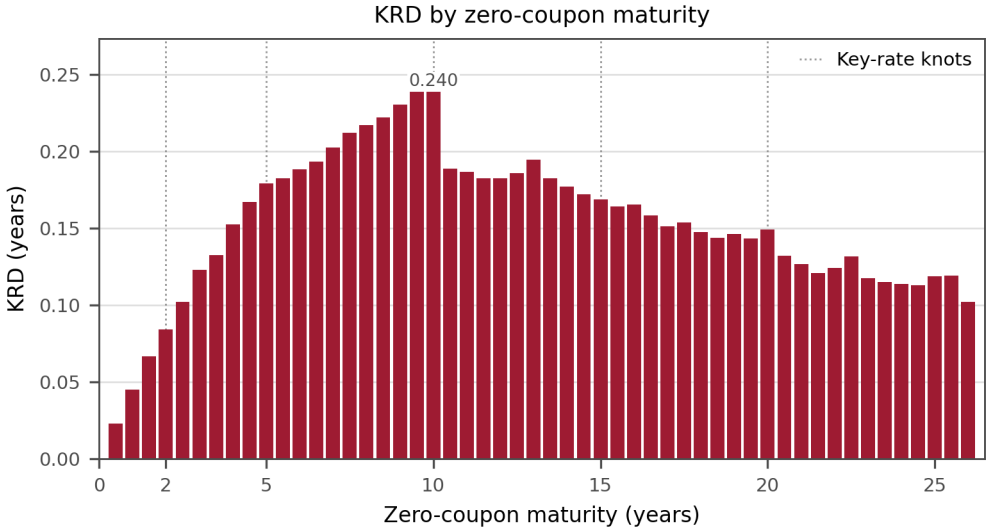



Figure 5: KRD to each individual zero-coupon yield (+50 bp). The 52 bars sum to 7.981 years, the portfolio's effective duration.

**Additivity check.** The 52 KRDs sum to 7.9809 years, exactly the effective duration of a ±50 bp parallel shift: a parallel shift is the sum of the 52 single-date shocks, so this check needs no curve model and is free of cross terms.

**Shape of the profile.** The exposure at date $t$ is roughly $t\times$(PV of the portfolio cash flow at $t$)/NAV. It rises with $t$ to a peak of 0.24y at 10y, then *drops abruptly* at 10.5y. Bonds maturing from Feb-2014 onwards carry high coupons (5.25–13.25%) and mostly trade far above par (up to

174), so an equal *dollar* allocation buys less face — hence less principal — at those dates. Beyond 10.5y the exposure declines slowly as discounting erodes the PV of distant cash flows.

**(b) The five key rates (aggregated view).** We apply the same +50 bp re-fit shock to each knot in turn.

| | Spline re-fit (Q2.1 model) | | | Tent (comparison) | |
|---|---|---|---|---|---|
| Key rate | P&L ($) | KRD (yrs) | DV01 ($) | P&L ($) | KRD (yrs) |
| 2Y | −1,820 | 0.367 | 36.7 | −2,640 | 0.531 |
| 5Y | −7,594 | 1.541 | 154.1 | −7,156 | 1.446 |
| 10Y | −10,813 | 2.210 | 221.0 | −10,123 | 2.061 |
| 15Y | −3,496 | 0.810 | 81.0 | −8,236 | 1.690 |
| 20Y | −13,980 | 3.064 | 306.4 | −10,717 | 2.252 |
| Sum | −37,704 | 7.992 | 799.2 | −38,872 | 7.979 |

Table 6: Key-rate exposures of the $1M equally-weighted portfolio (+50 bp shock at each knot). The parallel-shift effective duration is 7.981 years. KRD and DV01 are central differences; the P&L column is one-sided, so P&L $\neq -50\times$DV01.

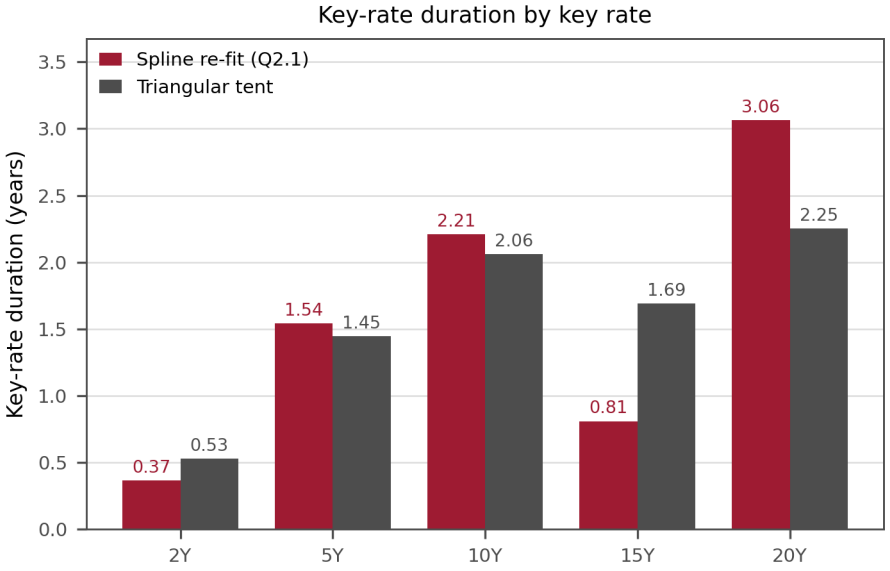



Figure 6: Key-rate duration profile (years): spline re-fit vs. triangular tent.

**Consistency check.** Both families of shocks sum to a parallel shift, so the KRDs should add up to the effective duration. They do: 7.992y (spline) and 7.979y (tent), versus 7.981y for a ±50 bp parallel shift. Summing the *one-sided* dollar P&Ls gives −$37,704 versus −$38,594 for the parallel shock. The gap comes from convexity and cross terms, which are largest for the spline because its extrapolated shocks reach about 100 bp. The flat-tail family introduced below, whose shocks never exceed 50 bp, sums to −$38,718 against −$38,594 for the parallel shift. This is why we report central-difference KRDs.

**Robustness: the 15Y/20Y split is an extrapolation effect.** The Q2.1 spline is pinned by data only up to 20Y; beyond that, the end cubic amplifies the effect of a 50 bp bump at 20Y to +103 bp at 26Y, and the effect of a 50 bp bump at 15Y to −50 bp at 24.5Y (−44 bp at 26Y). That is what moves duration between the two buckets:

| | 2Y | 5Y | 10Y | 15Y | 20Y | Sum |
|---|---|---|---|---|---|---|
| End cubic (reported) | 0.367 | 1.541 | 2.210 | 0.810 | 3.064 | 7.992 |
| Flat beyond 20Y | 0.359 | 1.570 | 2.115 | 1.848 | 2.087 | 7.980 |

If the tail is instead held flat beyond 20Y — the 20Y shock stays at +50 bp, the other four are zero there, and the end cubic is kept below 2Y — the split moves to 1.85y (15Y) and 2.09y (20Y), and the five KRDs still sum to 7.98y (7.980y vs. the 7.981y effective duration; the gap is the ±50 bp finite difference): under that convention the five shocks still sum to +50 bp at every maturity, so the decomposition is complete. Both conventions are complete; they disagree only on where the extrapolated tail is allocated (the end cubic gives 20Y 3.06y and 15Y 0.81y, the flat tail 2.09y and 1.85y). We report the end-cubic profile because Q4.1 asks for the shock on the curve estimated in Q2.1, and present the 15Y/20Y split as model risk rather than as a tradable exposure: under the flat-tail convention the 15Y leg of the illustrative hedge below flips from +$19k to −$153k.

**Interpretation.**

- All exposures are positive durations (negative P&L), as expected for a long-only portfolio.
- With the tent, the profile is smooth (0.53 / 1.45 / 2.06 / 1.69 / 2.25y) and follows the cash-flow intuition; the 20Y bucket is the largest because it also absorbs every maturity from 20 to 26y.
- With the spline re-fit, the ripples of Q4.1 redistribute risk. The 2Y KRD falls to 0.37y (vs. 0.53y): the 2Y bump pushes 6–9y yields *down* by up to 8.4 bp (the shock is zero at the 5Y and 10Y knots), worth +$1,063 on the 5–10y cash flows, which more than offsets the extra −$219 caused by its larger short-end shock (+79 bp at 0.5y vs. +50 bp for the tent). The 15Y/20Y split moves as explained above. The total is unchanged, which is the point.
- **Practical implication.** The choice of curve model is itself a source of hedging risk. A hedge sized on spline KRDs would put far more weight on 20Y and far less on 15Y than a hedge sized on tent KRDs, even though both imply the same total duration. Under the flat-tail convention the profile (1.85/2.09y at 15Y/20Y) sits close to the tent (1.69/2.25y), so most of the discrepancy is the end cubic, not the key-rate method.

**Illustrative KRD hedge (beyond the question).** Neutralising the five spline key-rate DV01s with the five bonds maturing exactly at the key dates requires face amounts of short $93.6k (Aug-2005), short $199.5k (Aug-2008), short $211.5k (Aug-2013), *long* $19.4k (Aug-2018) and short $445.6k (Aug-2023) — a net face position of −$930.9k (−$930.8k from the rounded legs), corresponding to a hedge market value of −$968k at those bonds' market prices (the funding leg is not modelled). The +50 bp parallel P&L falls from −$38,594 to +$57; the individual key-rate residuals are at most $606 (20Y), from second-order convexity. Under the flat-tail convention all five legs are short in face amount ($79.0k, $190.7k, $160.6k, $153.3k and $306.1k), so the 15Y leg flips from +$19.4k to −$153.3k and the 20Y leg drops from −$445.6k to −$306.1k.

# 2 Part II — Python Labs: US Treasury Yield Curve Reaction to Trump Tariff Threats

### Data, event set and conventions

**Data.** Daily US Treasury constant-maturity yields for ten maturities, from 3 months to 30 years (FRED series `DGS3MO` to `DGS30`), downloaded with the course helper `fred.py`. The sample used in Q1 runs from 2 January 2015 to 1 August 2025, i.e. 2,647 trading days. Yields are in percent; all results are reported in basis points (bp).

**Code.** A single run of `ga_project/06_event_study.py` reproduces every number, table and figure of this part (output files are prefixed `pl_`). The original course notebook is in `ga_project/python_labs/`.

**Event set.** The notebook's `trump_actions` and `country_responses` dictionaries contain 75 events: 36 Trump actions and 39 foreign responses (Python silently keeps only the last entry of each duplicated date key, so two listed events are lost). Mapping the events to trading days:

- 8 events fall after 1 August 2025, when the FRED download ends, and are dropped;
- 16 events fall on a weekend or a market holiday and are moved to the next trading day (e.g. the events of Saturday 5 and Sunday 6 April 2025 both map to Monday 7 April);
- 9 trading days carry two events. On 8 of them a Trump action and a foreign response share the same day (e.g. 2018-07-06, 2019-08-23, 2025-03-04); on the ninth, the 4 July 2025 holiday collapses onto the 7 July action.

The remaining 67 events therefore correspond to **58 distinct event days**, each counted once. All 58 have a full pre-event window; 51 have a full 30-day post-event window, because the last events of the sample are followed by fewer than 30 trading days of data.

**Methodology (as required by the assignment).** Let $t_1, \dots, t_N$ be the event days and $y_t$ a yield, or one of the three curve factors

$$\text{level} = y_{10}, \qquad \text{slope} = y_{10} - y_2, \qquad \text{curvature} = 2\,y_5 - y_2 - y_{10},$$

where $y_m$ is the $m$-year yield. For each horizon $k = -30, \dots, +30$ trading days, the abnormal change is

$$\mathrm{AC}(k) = \frac{1}{N}\sum_{i=1}^{N}\left(y_{t_i+k} - y_{t_i}\right) - \frac{1}{|\mathcal{B}_k|}\sum_{s\in\mathcal{B}_k}\left(y_{s+k} - y_s\right),$$

where $\mathcal{B}_k$ is the set of non-event days $s$ such that both $s$ and $s+k$ lie more than 30 trading days away from every event. The first term is the average change around the events; the second is the "normal" change over the same horizon.

**Reading the results.** Because $\mathrm{AC}(k)$ is measured relative to the event day, a *positive* value at $k < 0$ means that yields were *higher* before the event, i.e. they *fell* into it. To avoid this sign trap, every table reports three windows as the abnormal change in yield *over the window* (positive = yields rose):

- **pre-event** $[-30, 0]$: $y_t - y_{t-30}$, i.e. $-\mathrm{AC}(-30)$;
- **announcement** $[-1, +1]$: $y_{t+1} - y_{t-1}$, i.e. $\mathrm{AC}(+1) - \mathrm{AC}(-1)$;
- **post-event** $[0, +30]$: $y_{t+30} - y_t$, i.e. $\mathrm{AC}(+30)$.

**Assumptions.**

(A1) Business days are the trading days present in the FRED data; weekend and holiday events move to the next trading day.

(A2) The "reaction of targeted countries" (Q1.3) is measured on the *US* curve on the dates of the foreign responses, since only US yields are provided.

(A3) Q1 uses the notebook's event list, sample and baseline unchanged; all corrections are made in Q2.1.

(A4) To judge magnitudes we also report the *raw* average change around the events with its cross-sectional standard error (s.e.). These s.e. treat events as independent; since windows overlap, they overstate precision and are only a first indication.

## Q1.1 — Market reaction to Trump's announcements and actions

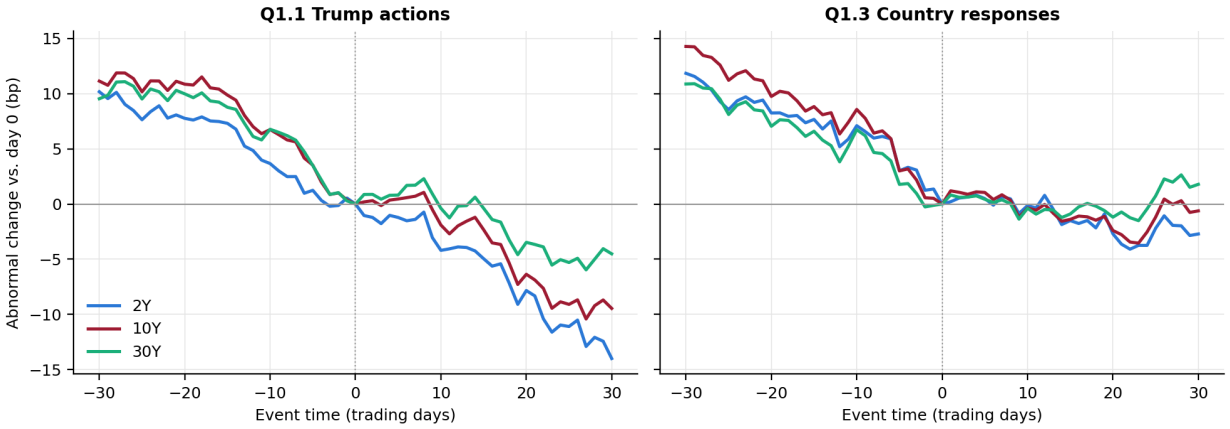



Figure 7: Abnormal change relative to the event day (bp), assignment methodology. Left: Trump actions (Q1.1). Right: foreign responses (Q1.3). A line falling from left to right means that yields fell through the window.

| | 3M | 2Y | 5Y | 10Y | 30Y |
|---|---|---|---|---|---|
| *Panel A. Abnormal change (assignment methodology), bp* | | | | | |
| Pre-event $[-30, 0]$ | −6.4 | −10.2 | −11.6 | −11.1 | −9.5 |
| Announcement $[-1, +1]$ | −0.5 | −1.6 | −0.8 | −0.2 | +0.6 |
| Post-event $[0, +30]$ | −7.1 | −14.0 | −13.6 | −9.5 | −4.5 |
| *Panel B. Post-event window decomposed: abnormal = raw − baseline, bp* | | | | | |
| Raw change around events | +1.4 | −6.8 | −7.4 | −3.5 | +1.0 |
| (s.e.) | (3.1) | (5.3) | (5.5) | (5.2) | (4.8) |
| Non-event baseline | +8.5 | +7.2 | +6.2 | +5.9 | +5.5 |

Table 7: Trump actions: 32 event days in the pre-event and announcement windows, 28 in the post-event window. Panel B may not add up exactly because of rounding.

**Discussion.**

- **No announcement effect.** Over $[-1, +1]$ the curve barely moves: −1.6 bp on the 2Y and −0.2 bp on the 10Y, against raw s.e. of about 1.2–1.5 bp. Nothing is distinguishable from zero.
- **Large abnormal declines on both sides of the event.** Taken at face value, Panel A says that yields fall by 10–12 bp (2Y to 10Y) in the 30 days *before* the events and by a further 5–14 bp *after* them, most at the front end. The textbook reading is a negative growth shock: flight to quality into Treasuries and higher odds of Fed cuts, which lowers the 2Y most.
- **But most of it is the baseline.** Panel B shows that the raw change after the events is small and statistically zero: −3.5 bp on the 10Y (s.e. 5.2) and +1.0 bp on the 30Y. The "abnormal" declines come mainly from subtracting a non-event baseline of +5.5 to +8.5 bp per 30 days. That baseline is computed on days far from any event, i.e. mostly 2015–16 and 2021–24, a period dominated by the 2022–23 hiking cycle, when yields were rising.
- **The shape confirms it.** A reaction to news would be a jump around day 0; Figure 7 instead shows a smooth 60-day ramp that starts a month before the announcements. This is the signature of a trend difference between event periods (2018–19, 2025) and the baseline period, not of a tariff effect.

**Answer.** With the required methodology, Treasury yields appear to fall around Trump's tariff actions (about −10 to −14 bp at the 2Y–5Y over 30 days). However, there is no announcement effect, and the post-event decline is driven by the choice of baseline rather than by the events. Q2.1 tests this directly.

## Q1.2 — Level, slope and curvature

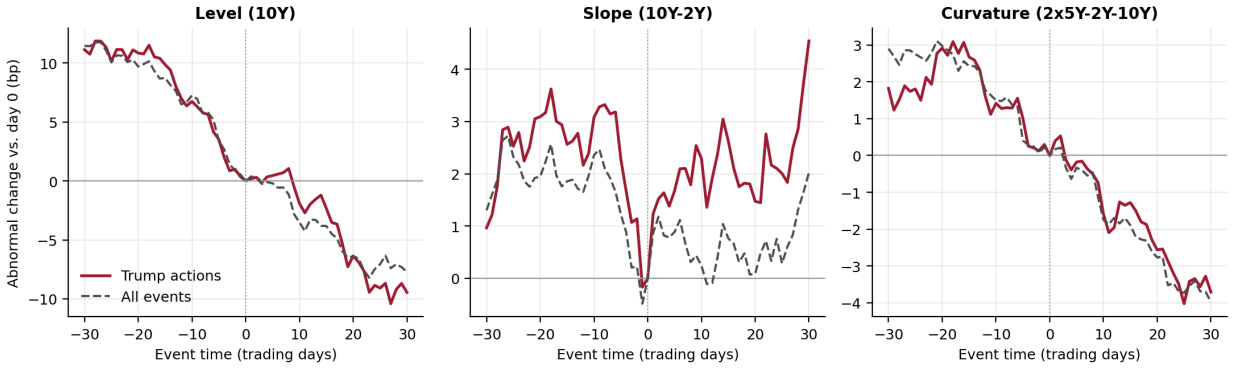



Figure 8: Abnormal change of level, slope and curvature (bp): Trump actions (red) vs. all events (dashed).

| | Trump actions | | | All events | | |
|---|---|---|---|---|---|---|
| Abnormal change, bp | Level | Slope | Curv. | Level | Slope | Curv. |
| Pre-event $[-30, 0]$ | $-11.1$ | $-1.0$ | $-1.8$ | $-11.5$ | $-1.3$ | $-2.9$ |
| Announcement $[-1, +1]$ | $-0.2$ | $+1.4$ | $+0.1$ | $-0.2$ | $+1.3$ | $-0.1$ |
| Post-event $[0, +30]$ | $-9.5$ | $+4.5$ | $-3.7$ | $-7.7$ | $+2.0$ | $-4.0$ |
| *Raw post-event change (s.e.)* | $-3.5$ (5.2) | $+3.3$ (1.9) | $-4.4$ (1.2) | | | |

Table 8: Level (= 10Y), slope (= 10Y − 2Y) and curvature (= 2×5Y − 2Y − 10Y).

**Discussion.**

- **Level** reproduces Q1.1 by construction, since it is the 10Y. The level effect is therefore subject to the same baseline caveat: raw $-3.5$ bp, s.e. 5.2. A level factor based on all maturities (first principal component) is used in Q2.1.
- **Slope.** After Trump actions the curve *steepens*: $+1.4$ bp over $[-1, +1]$ and $+4.5$ bp over 30 days, while the level falls. This is a *bull steepening*: the front end, anchored to expected policy rates, rallies more than the 10Y, which also carries tariff-driven inflation risk and a term premium. Unlike the level, the slope is barely affected by the baseline ($-1.3$ bp): the raw steepening is $+3.3$ bp (s.e. 1.9). It is partly offset by a flattening in the run-up ($-1.0$ bp abnormal), so the evidence is weak but this is the most event-like pattern in Q1.
- **Curvature** falls after the events ($-3.7$ bp): the 5Y belly rallies more than the average of the 2Y and 10Y wings. This would be consistent with cuts priced over the next 2–5 years. However, curvature also falls *before* the events (raw $-2.5$ bp) and after foreign responses (raw $-4.5$ bp). That points to a downward trend in curvature during the event periods rather than a reaction to tariff news.
- **All events vs. Trump actions.** Level and slope effects are weaker when foreign responses are included (post-event slope $+2.0$ vs. $+4.5$ bp), while curvature is unchanged, consistent with the trend interpretation.

## Q1.3 — Market reaction to the targeted countries' responses

**Discussion.**

- **Before the responses** yields fall by 11–15 bp (2Y to 30Y). This is largely mechanical: a response typically comes a few days after the US measure it answers, so its pre-event window contains the US action itself. In eight cases the two fall on the same trading day.

| | 3M | 2Y | 5Y | 10Y | 30Y | Slope | Curv. |
|---|---|---|---|---|---|---|---|
| *Panel A. Abnormal change (assignment methodology), bp* | | | | | | | |
| Pre-event $[-30, 0]$ | −2.6 | −11.8 | −15.3 | −14.3 | −10.9 | −2.4 | −4.4 |
| Announcement $[-1, +1]$ | +0.1 | −1.1 | −0.2 | +0.7 | +0.9 | +1.8 | 0.0 |
| Post-event $[0, +30]$ | −3.2 | −2.7 | −3.7 | −0.6 | +1.8 | +2.1 | −4.1 |
| *Panel B. Raw post-event change, bp* | | | | | | | |
| Raw change | +4.2 | +4.1 | +2.2 | +4.8 | +6.5 | +0.6 | −4.5 |
| (s.e.) | (2.3) | (3.7) | (3.8) | (3.8) | (3.7) | (2.2) | (1.0) |

Table 9: Foreign responses: 34 event days in the pre-event window, 33 in the announcement window, 31 in the post-event window.

- **At the announcement** nothing moves beyond noise (at most +0.9 bp on yields, +1.8 bp on the slope).
- **After the responses**, the abnormal change is close to zero (10Y −0.6 bp). In raw terms, however, yields *rise* by 2–7 bp, the 30Y the most (+6.5 bp, s.e. 3.7). These rises are absorbed by the rising baseline. So "no abnormal reaction" means that yields behaved like in an average non-event month, not that they were flat.
- **Economic reading.** Retaliation has two opposite effects on US yields. It hurts US exporters, which is growth-negative and lowers yields. It also raises import prices and threatens foreign demand for Treasuries, which pushes up inflation expectations and the term premium. The mild rise of the long end relative to the front end (slope +1.8 bp at the announcement) suggests that the second channel slightly dominates. The evidence, however, is weak and concentrated in 2025 (Q1.4).

## Q1.4 — First vs. second mandate

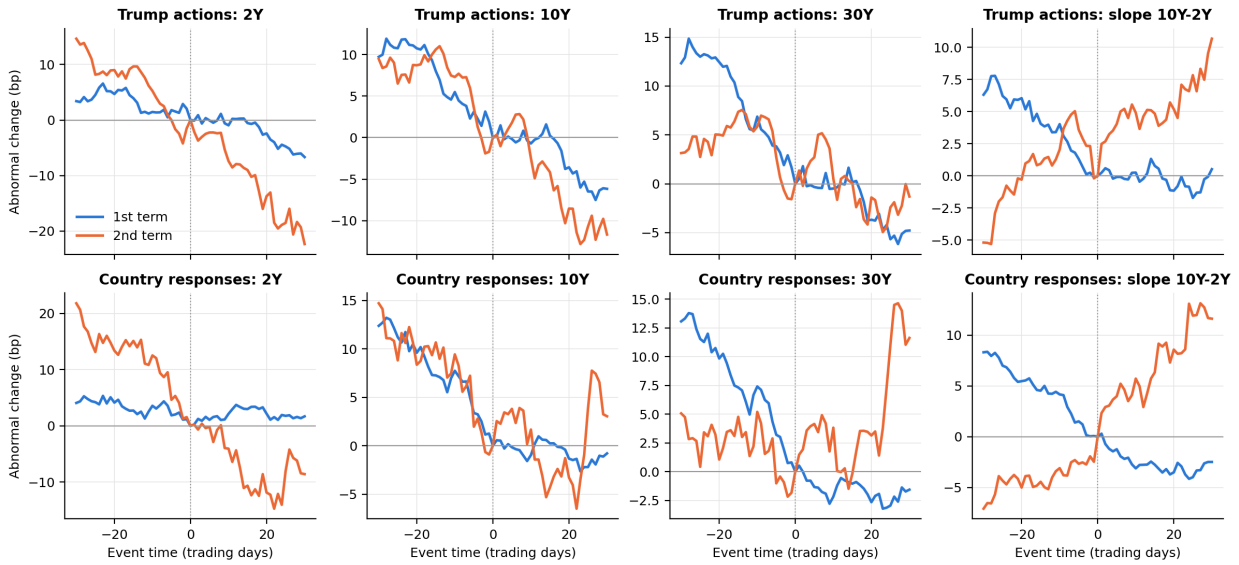



Figure 9: Abnormal change by mandate: first term (2017–20, blue) vs. second term (2025, orange). Top: Trump actions. Bottom: foreign responses.

**Discussion.** The two mandates look very different, but for a different reason than Panel A alone suggests.

- **First term (events in 2018–19): no measurable reaction.** After Trump actions the abnormal changes suggest a parallel rally of 5–7 bp. In raw terms, however, the 2Y, 10Y and 30Y are all within 1 bp of zero (s.e. about 6–7 bp): the apparent rally is entirely the baseline.

| | | Pre-event $[-30, 0]$, bp | | | | Post-event $[0, +30]$, bp | | | |
|---|---|---|---|---|---|---|---|---|---|
| Group | $n$ | 2Y | 10Y | 30Y | Slope | 2Y | 10Y | 30Y | Slope |
| *Abnormal change (assignment methodology)* | | | | | | | | | |
| Actions, 1st term | 18/18 | −3.4 | −9.7 | −12.3 | −6.3 | −6.7 | −6.2 | −4.8 | +0.5 |
| Actions, 2nd term | 14/10 | −14.7 | −9.5 | −3.1 | +5.2 | −22.4 | −11.7 | −1.3 | +10.7 |
| Responses, 1st term | 22/22 | −4.1 | −12.4 | −13.1 | −8.3 | +1.7 | −0.8 | −1.6 | −2.5 |
| Responses, 2nd term | 12/9 | −21.8 | −14.7 | −5.1 | +7.1 | −8.6 | +3.0 | +11.6 | +11.6 |
| *Raw post-event change (s.e.)* | | | | | | | | | |
| Actions, 1st term | | | | | | −0.7 (6.7) | −0.7 (6.8) | +0.8 (6.1) | 0.0 (2.2) |
| Actions, 2nd term | | | | | | −17.9 (7.9) | −8.7 (8.5) | +1.2 (8.2) | +9.2 (3.0) |
| Responses, 2nd term | | | | | | −4.1 (7.8) | +6.0 (7.4) | +14.1 (7.4) | +10.1 (4.5) |

Table 10: Yield change by mandate. Actions = Trump actions; $n$ = event days in the pre-/post-event windows.

The only clear first-term feature is a *flattening into the events* (abnormal slope over the pre-event window: −6.3 bp for actions, −8.3 bp for responses). This matches the general flattening of the 2018 hiking cycle more than tariff news. Tariffs were narrower (mainly China, plus steel and aluminium), inflation was low, and the Fed's reaction function dominated the curve.

- **Second term (2025): a steepening.** Here the raw numbers move:
  - After Trump actions, the 2Y falls by 18 bp while the 30Y does not move, giving a *bull steepening* (slope +9.2 bp, s.e. 3.0): the front end prices Fed cuts against weaker growth.
  - After foreign responses, the 30Y *rises* by 14 bp (slope +10.1 bp), a *bear steepening*.

  The long end no longer acted as a safe haven. Universal tariffs are inflationary, fiscal concerns were acute (Moody's downgrade in May, the One Big Beautiful Bill Act in July), and markets questioned foreign demand for Treasuries during the April "sell America" episode. The shock was absorbed by the term premium rather than by lower long yields.
- **Reliability.** The second-term samples are small: 10 and 9 event days in the post-event window. They are also concentrated between February and July 2025, so their 30-day windows overlap and represent only a handful of independent episodes. The s.e. above are therefore too small, and the differences between mandates are indicative only. Q2.1 tests which of them survive proper inference.

## Q2.1 — Limitations, improvements, and how they change Q1

### (a) Limitations of the design

1. **Biased baseline (regime effect).** The "normal" change is estimated on non-event days, which fall mostly in 2015–16 and 2021–24, a period dominated by the 2022–23 hiking cycle. The 10Y non-event baseline therefore rises by +5.9 bp per 30 days (Table 7), while the average 30-day change was −3.6 bp in 2017–20 and −5.8 bp in 2025 (Table 11). Subtracting a rising baseline from falling-rate periods mechanically creates a negative "abnormal" change. Q1 shows that this explains most of the post-event declines (raw vs. abnormal changes in Tables 7 and 9).
2. **Overlapping windows.** Events come in bursts (e.g. 14 events between 2025-02-01 and 2025-04-10). The 30-day windows therefore overlap and contain each other's events: observations are not independent, and each long window mixes many news items.
3. **No valid inference.** The assignment's design produces averages without a test. The naive standard errors added in Q1 treat overlapping windows as independent, so they overstate precision. With 61 horizons and several series, some "large" numbers are bound to appear by chance.
4. **Event-list quality.**
   - Two entries are silently lost to duplicated dictionary keys (2025-08-06 and 2025-03-04).

- Some entries are not shocks: "no retaliation" announcements, a status quo, and a US court ruling listed as a foreign response.
- Escalations and de-escalations (Phase One deal, delays, exemptions) are averaged together, so their effects offset each other.
- Major milestones are missing: the 90-day pause (2025-04-09), the US–China Geneva agreement (2025-05-12) and the US–EU deal (2025-07-27).
- Some descriptions are inaccurate (145% was the US rate on China).

5. **Sample truncation.** The download stops on 2025-08-01: eight August-2025 events are dropped and late-July windows are cut short.
6. **Timing.** Several announcements came after the market close ("Liberation Day" on 2025-04-02 was announced after 4pm ET) or on weekends, so day 0 is not always the reaction day (Figure 11).
7. **Confounding news.** Some event days coincide with FOMC decisions, and Trump actions and foreign responses share the same trading day eight times in the notebook list.
8. **Measurement.** Level = 10Y duplicates Q1.1. Constant-maturity par yields are used rather than zero rates, and daily closes cannot isolate an announcement from other news on the same day.

| Period | 2015–16 | 2017–20 (1st term) | 2021–24 | 2025 (2nd term) |
|---|---|---|---|---|
| Mean 30-day 10Y change (bp) | +3.3 | −3.6 | +9.7 | −5.8 |

Table 11: Average 30-day change of the 10Y yield by period. The 10Y non-event baseline of the assignment (+5.9 bp, Table 7) is dominated by the 2021–24 hiking cycle.

### (b) Improvements implemented

All improvements are in `06_event_study.py`, function `part_q2`.

1. **Cleaned and signed event list.**
   - Duplicated keys merged and six non-events removed (list in the code).
   - Each event signed as escalation (+1) or de-escalation (−1).
   - Three missing de-escalation milestones added.
   - Result: 72 events, of which 63 escalations and 9 de-escalations.
2. **Extended sample** to 2025-10-31, so that every event has a full window.
3. **Short windows** $[-1, +1]$ and $[0, +5]$ to capture the announcement effect. The $[-1, +1]$ window also covers after-close announcements, whose reaction occurs on day +1.
4. **Baseline from the same mandate.** The expected change is the average change over windows of the same length that start on ordinary days of the same mandate, i.e. days with no event within ±1 day (913 days in the first term, 127 in the second). Each event is thus compared with a typical window of its own rate regime, not with a 2015–2025 average.
5. **Placebo inference.** For each group we draw 5,000 sets of pseudo-event dates from the same pools, keeping the number of events per mandate. The two-sided $p$-value is the share of placebo averages at least as extreme as the actual one.
   - A stricter pool (no event within ±5 days) leaves only 56 days in 2025. These are unusually calm days, so the placebo distribution becomes too narrow and the $p$-values too optimistic.
   - With that pool, 6 results lose significance at 5% and 5 gain it, but none of the conclusions below changes (`pl_q2_placebo_sensitivity.csv`).

6. **De-clustering.** The 30-day window uses only the first event of each cluster (no other event in the previous 5 trading days).
7. **Robustness.** We exclude events with a same-day foreign response and events within ±1 day of an FOMC decision, and we replace the 10Y by a PCA level (first principal component of daily changes). For 2025 we also drop April, the month of "Liberation Day" and of the Treasury basis-trade deleveraging (Table 13).

| | $[-1,+1]$ | | | | $[0,+5]$ | | | $[0,+30]$, cluster heads | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Group | 2Y | 10Y | 30Y | Slope | 10Y | 30Y | Slope | 2Y | 10Y | $n$ |
| Trump escalations | +1.2 | +1.5 | +2.1 | +0.3 | +1.8 | +2.0 | +2.0* | +6.6 | +6.1 | 31/18 |
| 1st term | −2.3 | −2.2 | −1.5 | +0.1 | −0.6 | −0.5 | −0.4 | +7.9 | +7.2 | 16/12 |
| 2nd term | +5.1** | +5.5** | +6.0** | +0.4 | +4.3 | +4.7* | +4.6** | +4.1 | +4.1 | 15/6 |
| Foreign escalations | +1.8 | +3.5** | +3.7** | +1.7* | +4.3** | +4.1** | +2.1* | +4.6 | +7.7 | 32/10 |
| 1st term | +0.2 | +0.4 | +0.7 | +0.2 | +0.6 | −0.4 | −1.4 | +9.5 | +7.7 | 20/7 |
| 2nd term | +4.6** | +8.8** | +8.8** | +4.2** | +10.5** | +11.6** | +7.7** | −6.9 | +7.6 | 12/3 |
| De-escalations (all) | +7.0** | +6.7** | +5.9** | −0.3 | +1.9 | +2.9 | +2.6 | −20.6** | −19.7** | 9/4 |

Table 12: Improved design: abnormal change in bp (positive = yields up) relative to ordinary windows of the same mandate, extended sample, cleaned events. Placebo $p$-values: $^{*}p < 0.10$, $^{**}p < 0.05$. $n$ = events in the short / 30-day windows.

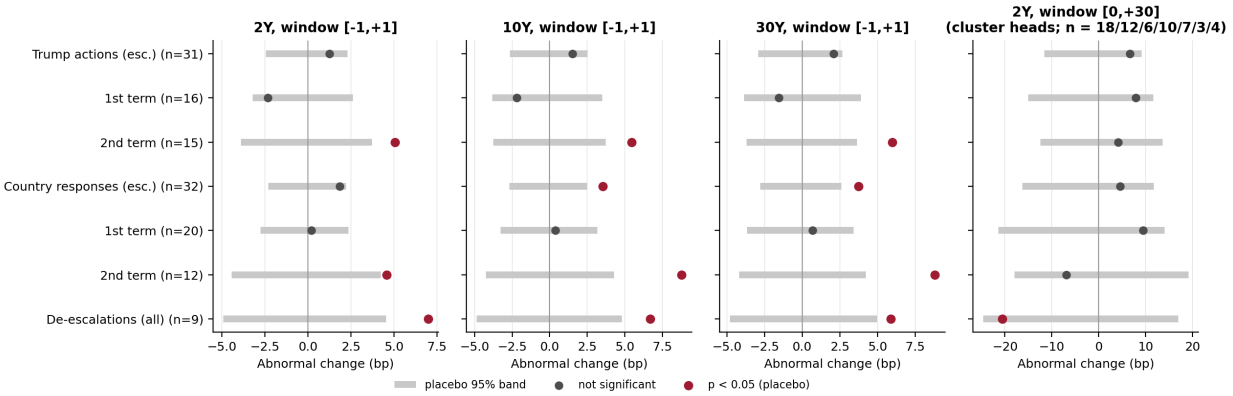



Figure 10: Improved estimates (dots) against the 95% range of the placebo distribution (grey bars). Red dots are significant at 5%.

| | | 10Y | | 30Y | |
|---|---|---|---|---|---|
| 2025 escalations, window $[-1,+1]$ | $n$ | bp | $p$ | bp | $p$ |
| Trump actions, all | 15 | +5.5 | 0.005 | +6.0 | 0.002 |
| excl. April 2025 | 13 | +5.7 | 0.006 | +5.7 | 0.005 |
| excl. same-day foreign response | 12 | +3.4 | 0.112 | +3.7 | 0.072 |
| Foreign responses, all | 12 | +8.8 | $< 0.001$ | +8.8 | $< 0.001$ |
| excl. April 2025 | 9 | +6.7 | 0.005 | +6.7 | 0.006 |
| excl. same-day Trump action | 9 | +7.1 | 0.004 | +6.8 | 0.005 |

Table 13: Robustness of the 2025 announcement effect (placebo $p$-values). No 2025 escalation falls within ±1 day of an FOMC decision, so that exclusion changes nothing.

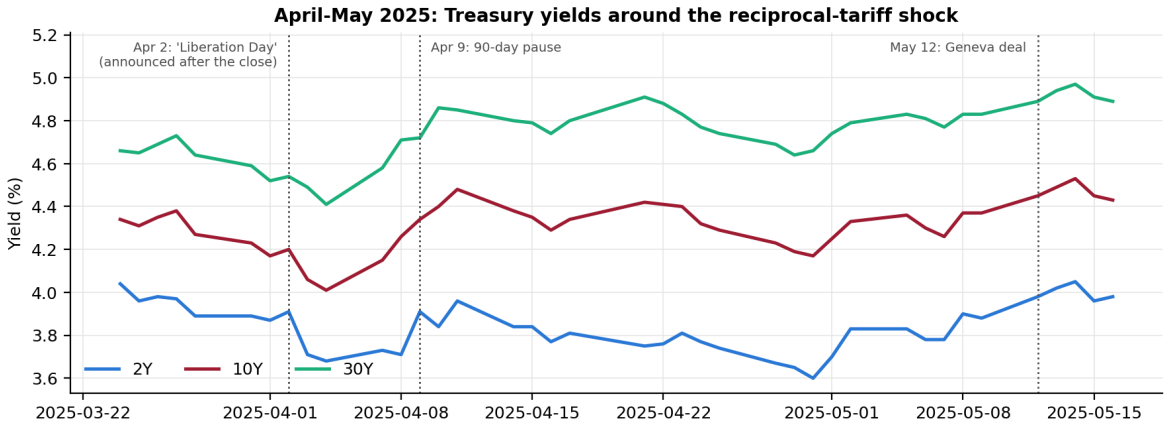



Figure 11: Daily yields around "Liberation Day". The announcement came after the close on 2 April, so the reaction is on 3 April (−20 bp 2Y, −14 bp 10Y). From 7 to 10 April the long end sold off (+45 bp on the 30Y between the 4 and 10 April closes) while tariffs were still escalating.

### (c) How the improvements change our answers to Q1

| | Assignment design (Q1) | Improved design (Q2) |
|---|---|---|
| Q1.1 | Abnormal declines of 10–14 bp after Trump actions, but no announcement effect and a raw change close to zero. | **Confirmed: no effect when both mandates are pooled.** Trump escalations have no significant announcement effect ($p \geq 0.13$; 1 of 20 robustness cells below 5%). The 30-day decline disappears (+6.6 bp on the 2Y, n.s.) once each event is compared with its own rate regime. |
| Q1.2 | Level down, slope up (+4.5 bp): an apparent bull steepening. | No level effect when pooled. A steepening over $[0, +5]$ (+2.0 bp, $p = 0.07$) comes entirely from 2025 (+4.6 bp$^{**}$), and it follows a *rise* in yields, not a front-end rally: a bear steepening. |
| Q1.3 | Abnormal change near zero after responses, while raw 10Y and 30Y yields rise by 5–7 bp. | **Significant rise in long yields**: 10Y +3.5$^{**}$ and 30Y +3.7$^{**}$ at the announcement, +4.1 to +4.3$^{**}$ over five days. The effect is entirely due to 2025 (10Y and 30Y +8.8$^{**}$, slope +4.2$^{**}$) and survives dropping April. |
| Q1.4 | 1st term: no raw reaction. 2nd term: front-end decline and steepening. | **1st term: no effect at any horizon. 2nd term: yields rise** by 5–6 bp across the curve on Trump escalation days and by 9 bp at the long end after retaliation; the curve then steepens. The 30-day front-end decline of 2025 is not event-specific: it is the general 2025 trend. Treasuries did not act as a safe haven. |
| New | — | **De-escalations also raise yields** (+6 to +7 bp, with an unchanged slope; e.g. +20 bp on the 2Y on 9 April 2025). This is a "risk-on" repricing of expected Fed cuts, which pooling escalations with de-escalations had hidden. |

Table 14: Conclusions before and after the methodological improvements.

**Overall.** The smooth 60-day patterns of Q1 are mostly a rate-regime effect plus clustering, not tariff news. Once these are removed, three findings remain:

1. First-term tariff news had no measurable effect on Treasury yields.
2. In 2025, escalation days, and above all foreign retaliation, *raised* yields and steepened the curve. The rise is largest at the long end, consistent with a rising term premium (inflation,

fiscal and foreign-demand concerns) rather than a flight to safety. It is not driven by April 2025 alone.

3. De-escalations raised yields roughly in parallel, as markets priced out Fed cuts. The *slope* separates the two channels: escalations steepen the curve, de-escalations do not.

**Remaining limitations.**

- **Multiple testing.** Table 12 reports 63 tests, 20 of them significant at 5%, against about 3 expected by chance. The significant cells, however, are concentrated in the 2025 groups, whose events fall in the same few months, so they are far from 20 independent findings.
- **Small samples.** The second-term groups have 12–15 events in the short windows and only 3–6 in the 30-day windows. The 30-day results, including the $-20$ bp for de-escalations ($n = 4$), are anecdotal.
- **Overlaps remain.** Removing same-day foreign responses weakens the 2025 Trump effect ($p = 0.07$–$0.11$), so part of it may be retaliation news released on the same day.
- **Daily data.** A cleaner design would use intraday Treasury futures in 30-minute windows around each announcement, and zero-coupon rather than par yields.
- **Foreign curves.** With the relevant data, foreign yields could measure the reaction in the targeted countries directly.

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